

- 9 (a)** The structure of a mitochondrion is related to its function in respiration.
- Figure 9.1 is a diagram of a mitochondrion.
- intermembrane DNA

Figure 9.1 is a diagram of a mitochondrion.

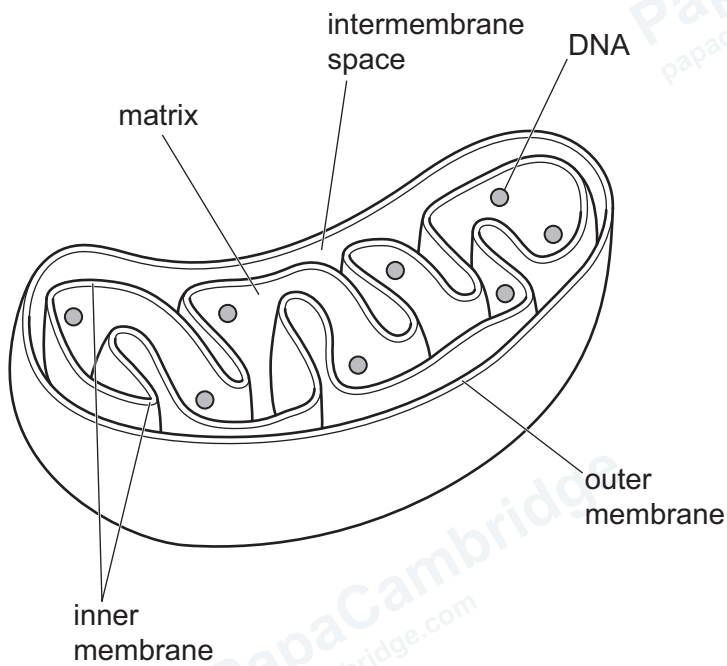


Figure 9.1

With reference to Figure 9.1, outline the relationship between the structure and function of a mitochondrion in respiration.

[4]

- (b) Table 9.1 shows the mean percentage of cell volume occupied by mitochondria in three different cell types.

Table 9.1

cell type	mean percentage of cell volume occupied by mitochondria
heart muscle cell	48
liver cell	17
red blood cell	0

Suggest why the three cell types shown in Table 9.1 have different mean percentages of cell volume occupied by mitochondria.

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..... [3]

- (c) Glycolysis occurs in the cytoplasm and produces pyruvate which, in aerobic conditions, can enter the mitochondrion.

Outline how a molecule of glucose is converted to molecules of pyruvate during glycolysis.

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..... [3]

[Total: 10]